HOMEWORK 1.2 · UNIT 1 · NC.8.NS.1

Number Classification

Targets: NS.1-4, NS.1-3, NS.1-1, NS.1-2, NS.1-7, NS.1-6, NS.1-8, NS.1-9

Name _____ Date _____ Block _____

Remember: Check every set, from the inside out: **Natural** {1, 2, 3, ...} → **Whole** (add 0) → **Integer** (add negatives) → **Rational** (add fractions and decimals that stop or repeat). **Irrational** is its own box. Example: $\frac{18}{3} = 6$ → Natural, Whole, Integer, Rational.

SCORE 3.0 Approaching · the basics

Targets: NS.1-4, NS.1-3, NS.1-1, NS.1-2

NS.1-4

1 -15 belongs to which sets?

- ☐ Natural
- ☐ Whole
- ☐ Integer
- ☐ Rational
- ☐ Irrational

Select all that apply.

NS.1-4

2 $\frac{20}{4}$ belongs to which sets?

Hint: Divide first.

- ☐ Natural
- ☐ Whole
- ☐ Integer
- ☐ Rational
- ☐ Irrational

Select all that apply.

NS.1-3

3 Circle one: $\sqrt{50}$

Rational Irrational

Why? _____

REVIEW 1.1 NS.1-1

4 Which decimal is **repeating**?

- (A) 0.25
- (B) 0.454545...
- (C) 3.1415926...
- (D) 0.5

REVIEW 1.1 NS.1-2

5 π is a rational number.

True False

SCORE 4.0 Proficient · on target

Targets: NS.1-7, NS.1-6

6 **Theo** made a mistake. Circle it. Explain it. Fix it. NS.1-7

THEO'S WORK

Sorting 2.5:
2.5 → Integer, Rational

7 Which numbers are **rational**? REVIEW 1.1 NS.1-6

- ☐ $\sqrt{64}$
- ☐ 0.121121112...
- ☐ -9
- ☐ $\frac{7}{8}$
- ☐ π
- ☐ $0.\overline{6}$

Select all that apply.

8 Write one number that is rational but **not** an integer. NS.1-7

SCORE 5.0 Mastery · go deeper

Targets: NS.1-8, NS.1-9

9 Explain why every whole number is also an integer. Use the real number system diagram. NS.1-8

Every whole number is an integer because _____ .

10 Is 5π rational or irrational? Explain. NS.1-9

5π is _____ because _____ .